

Worksheet_s4094828_UG_Group_35

COSC3047 - Social Media and Network Analysis

Assignment 2 - Group 35 (UG)

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Project title: How Is Melbourne Liveability Discussed on YouTube? A Network and Topic Analysis of User Participation, Replies, and Liveability Themes

Project period: Week 8 to Week 12 (27 April to 25 May 2026)

1. Team project plan

Our goal was to build a legally accessible, reproducible social-media analysis project that satisfied both required components of the assignment: a meaningful network analysis and a meaningful NLP/text-analysis component. After evaluating several possible topics and data sources, we selected a curated YouTube dataset on Melbourne liveability because it supported participation edges, reply edges, and a substantial public comment corpus suitable for topic modelling.

Planned workflow:

- Week 8: confirm topic, finalise data source, and define the research question and analytic scope.
- Week 9: collect and clean the YouTube dataset; validate that the data supported both network construction and text analysis.
- Week 10: build the preprocessing and network-analysis workflow, including the user-video bipartite graph, reply graph, and thresholded user-user projection.
- Week 11: complete topic modelling, evaluate candidate topic numbers, label topics, and prepare report-ready figures and tables.
- Week 12: extend the analysis with Louvain community detection and VADER sentiment, complete the report draft, check consistency between code and prose, package the repository, and prepare submission materials.

Milestones and planned responsibilities:

Week	Milestone	Primary tasks	Planned team contribution
Week 8	Topic and source selection	Select the final topic; confirm the legality and suitability of the data source; agree on research question and workflow.	Balanced planning and topic-scoping by both team members.
Week 9	Data collection and validation	Set up YouTube API collection scripts; gather curated video, comment, reply, and metadata files; check	Balanced contribution to data collection, validation, and debugging.

		data quality and structure.	
Week 10	Preprocessing and network analysis	Clean comments, create analysis subsets, construct edges, and build the bipartite, reply, and projection networks.	Balanced contribution to preprocessing, network construction, and result checking.
Week 11	Topic modelling and interpretation	Train and compare candidate LDA models; finalise topic labels; generate prevalence tables and report figures.	Balanced contribution to topic modelling, interpretation, and visualisation.
Week 12	Extensions, report, and submission packaging	Run Louvain and VADER extensions; refine the report; set up repository access; prepare final submission files.	Balanced contribution to extension analysis, report completion, and submission preparation.

2. Weekly timesheets

The timesheets below estimate our work across Weeks 8-12. The task split is presented as broadly balanced across the project, with both members contributing to topic definition, data work, analysis development, and report completion.

Week 8 (27 Apr - 3 May 2026)

Topic refinement, dataset-source evaluation, assignment planning, and initial workflow design.

Reviewed assignment requirements and sample reports; narrowed the project to Melbourne liveability and assessed whether YouTube could support both network analysis and NLP.

Team member	Hours	Main work completed
Raphael Urpani (s4094828)	5.0 hours	Topic scoping, source evaluation, project-planning discussion, workflow design, and suitability-form preparation.
Leon Zaritski (s4096125)	5.0 hours	Topic scoping, source evaluation, literature/background reading, project-planning discussion, and report-structure planning.

Week 9 (4 May - 10 May 2026)

YouTube API setup, collection-script adaptation, curated-video selection, dataset collection, debugging, and initial validation of raw outputs.

Team member	Hours	Main work completed
Raphael Urpani (s4094828)	6.0 hours	API setup, data collection, script debugging, raw-output inspection, and repository preparation.
Leon Zaritski (s4096125)	6.0 hours	Curated-video validation, data-quality checking, collection support, literature/background reading, and documentation notes for methods.

Week 10 (11 May - 17 May 2026)

Preprocessing pipeline completed; cleaned comments, analysis subset, video theme buckets, user-video edges, reply edges, and summary outputs produced. Initial network analysis completed.

Team member	Hours	Main work completed
Raphael Urpani (s4094828)	6.5 hours	Notebook work on cleaning, edge construction, preprocessing checks, and network-output interpretation.
Leon Zaritski (s4096125)	6.5 hours	Notebook work on preprocessing validation, network-result review, figure/table planning, and draft results framing.

Week 11 (18 May - 24 May 2026)

Topic-modelling workflow completed; compared $K = 4, 6, 8$; selected six-topic model; labelled topics using representative comments; produced prevalence figures and report tables.

Team member	Hours	Main work completed
Raphael Urpani (s4094828)	6.0 hours	Topic-model evaluation, figure review, topic-label checking, and integration of outputs into the project workflow.
Leon Zaritski (s4096125)	6.0 hours	Topic-model interpretation, report drafting, table/figure integration, and narrative development for topic-modelling results.

Week 12 (25 May 2026)

Extension notebook added for Louvain community detection and VADER sentiment; final report revision, GitHub repository setup, access preparation, and submission packaging.

Team member	Hours	Main work completed
Raphael Urpani (s4094828)	5.5 hours	Repository setup, submission packaging, extension-result

		checking, consistency review between code and report, and final QA.
Leon Zaritski (s4096125)	5.5 hours	Extension notebook completion, final report drafting and editing, results integration, and final submission review.

3. Workload summary

Across the five-week project window, we aimed for an even workload split. On the estimates above, both members contributed approximately 29.0 hours to the project. Responsibilities were differentiated by task emphasis rather than by overall workload: Raphael contributed more heavily to collection setup, preprocessing, network construction, repository setup, and final packaging, while Leon contributed more heavily to report drafting, extension analysis, and synthesis of results. In total, however, the workload was intended to remain broadly balanced.

4. Individual self-reflection (total length within 1 A4 page)

Raphael Urpani

My main contributions were in topic selection, YouTube data collection and debugging, preprocessing, network-analysis workflow development, repository setup, and final consistency checking between the code and the report. One of the main challenges was identifying a data source that was legally accessible, rich enough for both network analysis and NLP, and still realistic to collect within the project timeframe. The move to a curated YouTube dataset improved the quality and coherence of the project, but it also required careful design to ensure that the networks constructed were genuinely meaningful rather than superficial. I think we worked effectively overall: we were able to divide responsibilities across technical implementation, interpretation, and report writing while still checking each other's work. This project improved my understanding of how to connect data collection, preprocessing, network analysis, topic modelling, and reproducible reporting in a single workflow.

Leon Zaritski

My main contributions were in report drafting, interpretation of the results, extension analysis using Louvain community detection and VADER sentiment, and integration of figures and tables into a coherent research narrative. A challenge in this project was ensuring that the report remained tightly aligned to the outputs generated by the notebooks, especially once the project expanded beyond the initial core analysis. I think we collaborated well by maintaining a roughly even workload while taking responsibility for different parts of the pipeline. The strongest part of the project was the way the final report connected the participation structure, reply structure, topic model, community structure, and sentiment overlay into a single account of Melbourne liveability discourse. I learned a

great deal about the importance of clarity, reproducibility, and interpretive discipline in social-media analytics.